Exploratory Data Analysis: Store Type A

We are doing an Exploratory Data Analysis on Store Type "a". Here are the key questions we are answering to justify later modeling choices:

- What does the sales data look like over time: stable, trending, or volatile?
- Is there a repeating seasonal pattern (weekly/monthly)?
- What affects sales: promo, competition, and day of week?
- Is the series stationary to guide ARIMA differencing?

```r
library(tidyverse)
library(forecast)
library(tseries)
```

```r
agg <- read_csv("../data/processed/store_type_a_train_aggregate_daily.csv") %>%
  mutate(Date = as.Date(Date)) %>%
  arrange(Date)

store_daily <- read_csv("../data/processed/store_type_a_train_store_daily.csv") %>%
  mutate(Date = as.Date(Date))
```

Summary of data

Basic statistics of aggregated daily sales across all stores, gives us min, max, mean, median - helps us understand the range and distribution of sales

```r
summary(agg$Sales)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##       0 2994007 3563650 3360228 4459142 8511488
```

Date continuity check

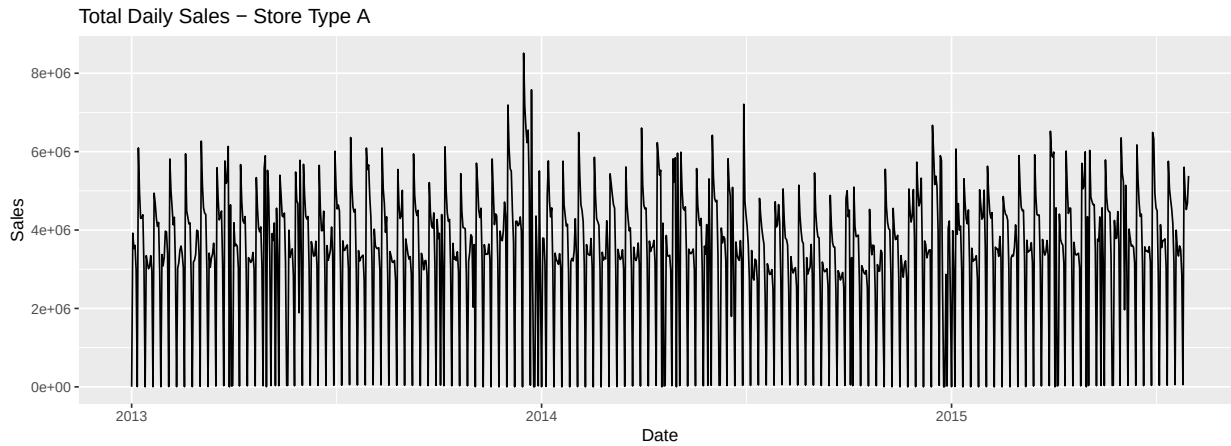
```r
date_span <- as.integer(max(agg$Date) - min(agg$Date)) + 1
missing_days <- date_span - nrow(agg)

tibble(
  first_date = min(agg$Date),
  last_date = max(agg$Date),
  expected_days = date_span,
  observed_days = nrow(agg),
  missing_days = missing_days
)
```

```
## # A tibble: 1 x 5
##   first_date last_date  expected_days observed_days missing_days
##   <date>     <date>             <dbl>         <int>        <dbl>
## 1 2013-01-01 2015-07-31           942           942            0
```

Plot of total daily sales for Store Type "a" over dates

We're trying to understand: is there a trend? Visible seasonality? Any weird outliers?

```r
ggplot(agg, aes(x = Date, y = Sales)) +
  geom_line() +
  labs(title = "Total Daily Sales - Store Type A", x = "Date", y = "Sales")
```



Some things that we can see here are that:

- There is a strong weekly seasonality, the regular repeating spikes and dips throughout the entire series
- There's no clear upward or downward trend, which means sales are roughly and relatively stable across 2013-2015
- There are some outlier spikes around early 2014, likely perhaps due to holidays or special events
- Some zero drops still visible, these could be during Sundays or holidays when all Type "a" stores were closed

Compare average sales on promo vs non-promo days

If mean sales are higher when PromoShare > 0, that supports including promo as an external regressor in ARMAX later.

```
promo_summary <- agg %>%
  mutate(PromoDay = if_else(PromoShare > 0, 1L, 0L)) %>%
  group_by(DayOfWeek, PromoDay) %>%
  summarise(
    mean_sales = mean(Sales, na.rm = TRUE),
    sd_sales = sd(Sales, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  )

promo_summary
```

```
## # A tibble: 12 x 5
##    DayOfWeek PromoDay mean_sales sd_sales     n
##        <dbl>    <int>      <dbl>    <dbl> <int>
##  1         1        0   3352902. 1294884.    62
##  2         1        1   5751046.  654095.    72
##  3         2        0   3394326.  653628.    63
##  4         2        1   4943941.  575095.    72
##  5         3        0   3244104.  712786.    63
##  6         3        1   4473674.  755051.    72
##  7         4        0   3042221. 1110822.    63
##  8         4        1   4272478.  917407.    72
##  9         5        0   3442823. 1009332.    66
## 10         5        1   4327555.  921347.    69
## 11         6        0   3125231.  443886.   134
```

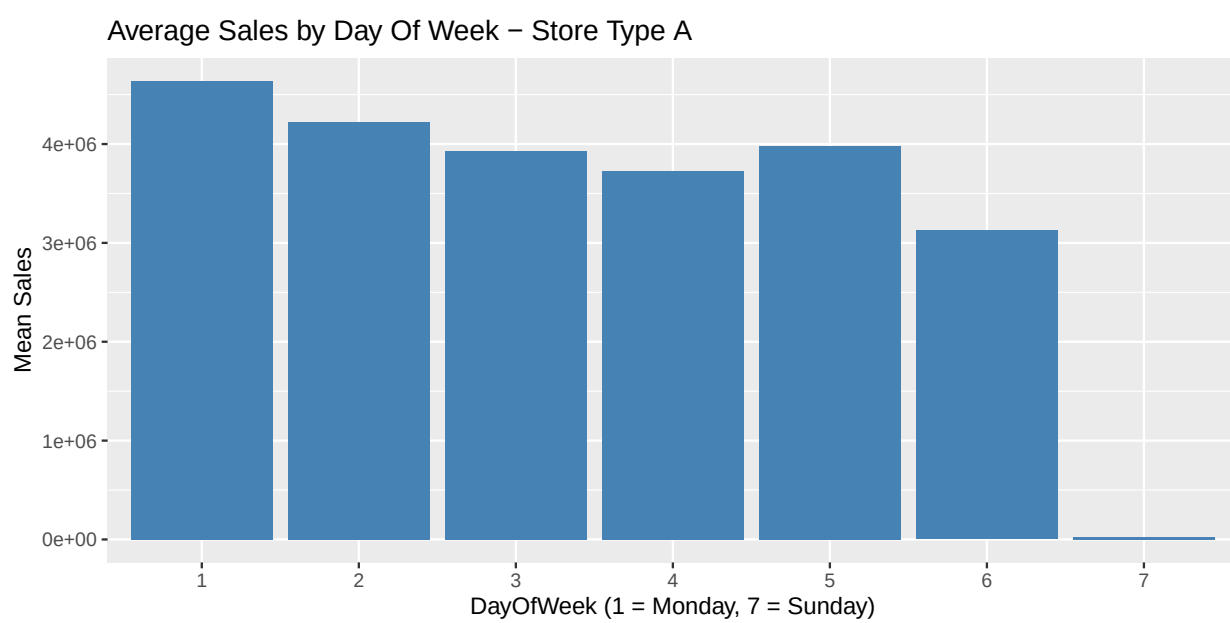
```
## 12          7         0     23905.   16795.   134
```

```r
promo_lm <- agg %>%
  filter(Sales > 0) %>%
  mutate(
    DayOfWeek = factor(DayOfWeek),
    Month = factor(Month)
  ) %>%
  lm(log1p(Sales) ~ PromoShare + OpenStores + DayOfWeek + Month, data = .)

coef(summary(promo_lm))["PromoShare", ]
```

```
##      Estimate    Std. Error       t value      Pr(>|t|)
## -4.667888e-03  4.520661e-04 -1.032568e+01  1.079745e-23
```

Day-of-week effect

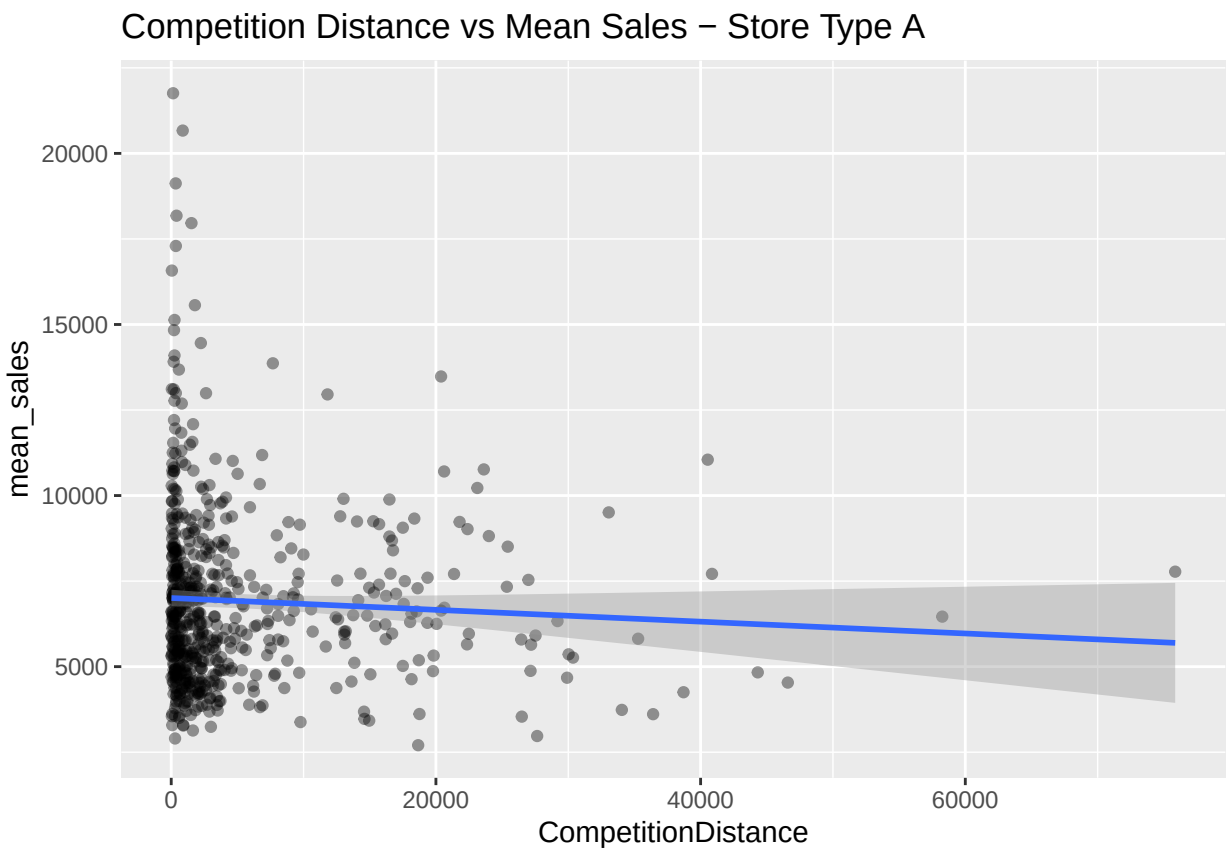
```r
dow_summary <- agg %>%
  filter(Sales > 0) %>%
  group_by(DayOfWeek) %>%
  summarise(mean_sales = mean(Sales, na.rm = TRUE), .groups = "drop")

ggplot(dow_summary, aes(x = factor(DayOfWeek), y = mean_sales)) +
  geom_col(fill = "steelblue") +
  labs(
    title = "Average Sales by Day Of Week - Store Type A",
    x = "DayOfWeek (1 = Monday, 7 = Sunday)",
    y = "Mean Sales"
  )
```



Scatter plot of competition distance vs mean sales for Store Type "a"

```r
store_daily %>%
  filter(Open == 1, Sales > 0, !is.na(CompetitionDistance)) %>%
```

```r
group_by(Store) %>%
summarise(mean_sales = mean(Sales), CompetitionDistance = first(CompetitionDistance)) %>%
ggplot(aes(x = CompetitionDistance, y = mean_sales)) +
geom_point(alpha = 0.4) +
geom_smooth(method = "lm") +
labs(title = "Competition Distance vs Mean Sales - Store Type A")
```

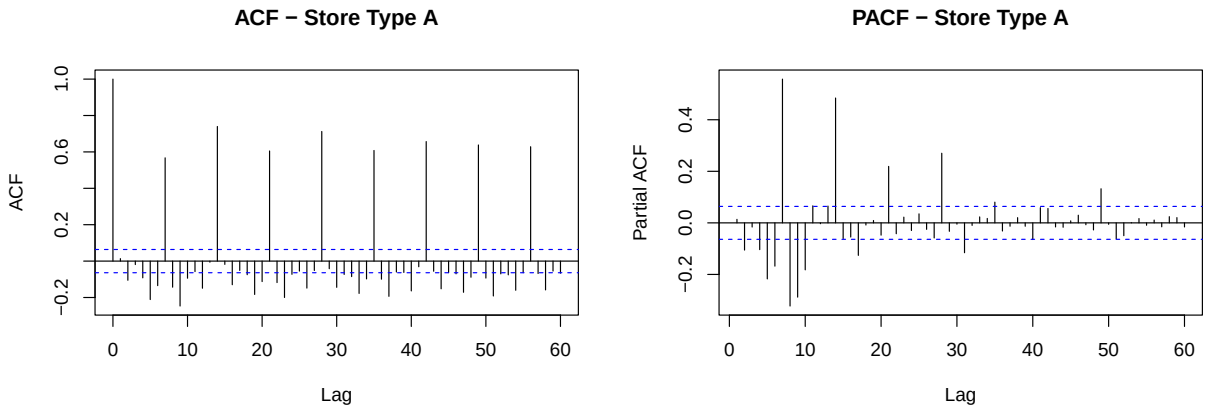


Since the line slopes downward, it means closer competitors hurt sales. This is counterintuitive but likely means those stores are located in busier, areas where competition is higher but foot traffic may be greater. The relationship is weak though, since the dots are very spread out, CompetitionDistance may only be a marginally useful explanatory variable in ARMAX.

ACF and PACF plots for Store Type "a"

ACF shows how correlated today's sales for Store Type "a" are with sales from previous days. Slow decay = series is not stationary (needs differencing in ARIMA). Spikes at lags 7, 14, 21 = confirms weekly seasonality

```r
sales_ts <- ts(agg$Sales, frequency = 7)
par(mfrow = c(1, 2))
acf(agg$Sales, lag.max = 60, main = "ACF - Store Type A")
pacf(agg$Sales, lag.max = 60, main = "PACF - Store Type A")
```




```r
par(mfrow = c(1, 1))
```

There are clear spikes at lags 7, 14, 21, 28 (multiples of 7), which confirms strong weekly seasonality every week, today's sales are highly correlated with sales exactly 7 days ago. This informs us that in our ARIMA model, we'd need a seasonal component with a period of 7.

PACF plot - shows the correlation at each lag after removing effects of shorter lags. Helps us identify the AR order (p) for our ARIMA model

Significant spikes at lags 7 and 14, with most other lags inside the blue dashed lines suggests the seasonal AR order is likely 1 or 2, which helps us later narrow down candidate ARIMA models later.

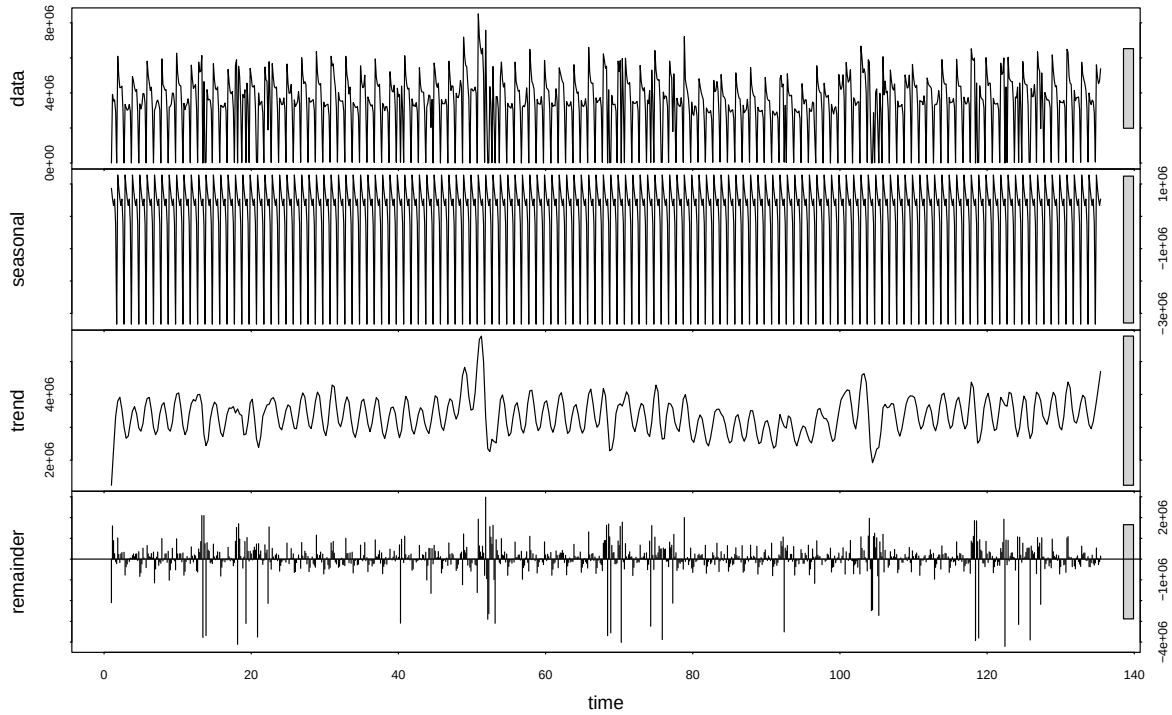
STL Decomposition

Splits series into three separate components:

- Trend = long term direction (going up or down over time?)
- Seasonal = repeating weekly pattern
- Remainder = what's left after removing trend and seasonality (random noise)

Great figure to include in the report as it visually proves seasonality exists

```r
stl_fit <- stl(sales_ts, s.window = "periodic")
plot(stl_fit)
```



Interpreting these four panels:

- Data: the raw series with all variation
- Trend: relatively flat overall with a noticeable outlier spike around early 2014, then settling back down. No strong long-term upward or downward direction
- Seasonal: there's very strong and consistent weekly pattern throughout the entire series, confirming weekly seasonality
- Remainder: mostly random noise around zero, with a few large spikes around early 2014 corresponding to perhaps unusual days like holidays

Augmented Dickey-Fuller and KPSS test

Augmented Dickey-Fuller test tests if series is stationary

- $p < 0.05$ = stationary (good, no differencing needed)
- $p > 0.05$ = not stationary (need d=1 in ARIMA)

KPSS test is another stationarity test, interpreted opposite to ADF

- $p > 0.05$ = stationary
- $p < 0.05$ = not stationary

```
adf_result <- adf.test(agg$Sales)
kpss_result <- kpss.test(agg$Sales)

adf_result
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  agg$Sales
```

```
## Dickey-Fuller = -15.968, Lag order = 9, p-value = 0.01
## alternative hypothesis: stationary
```

```
kpss_result
```

```
##
##  KPSS Test for Level Stationarity
##
## data:  agg$Sales
## KPSS Level = 0.12577, Truncation lag parameter = 7, p-value = 0.1
```

ADF test says p-value = 0.01 (less than 0.05), hence series is stationary

KPSS test says p-value = 0.1 (greater than 0.05), hence series is stationary

Since both tests agree the series is stationary, we likely won't need differencing (d=0) in the ARIMA model.

EDA notes for modeling

- Strong weekly behavior (period 7) supports seasonal ARIMA terms.
- Promo share, open stores, and day-of-week effects suggest useful external regressors for ARMAX.
- Stationarity test outcomes guide whether differencing is required.